

Problem

Design a circuit to amplify the difference between two inputs by 2.5.

- (a) Use only one op amp.
- (b) Use two op amps.

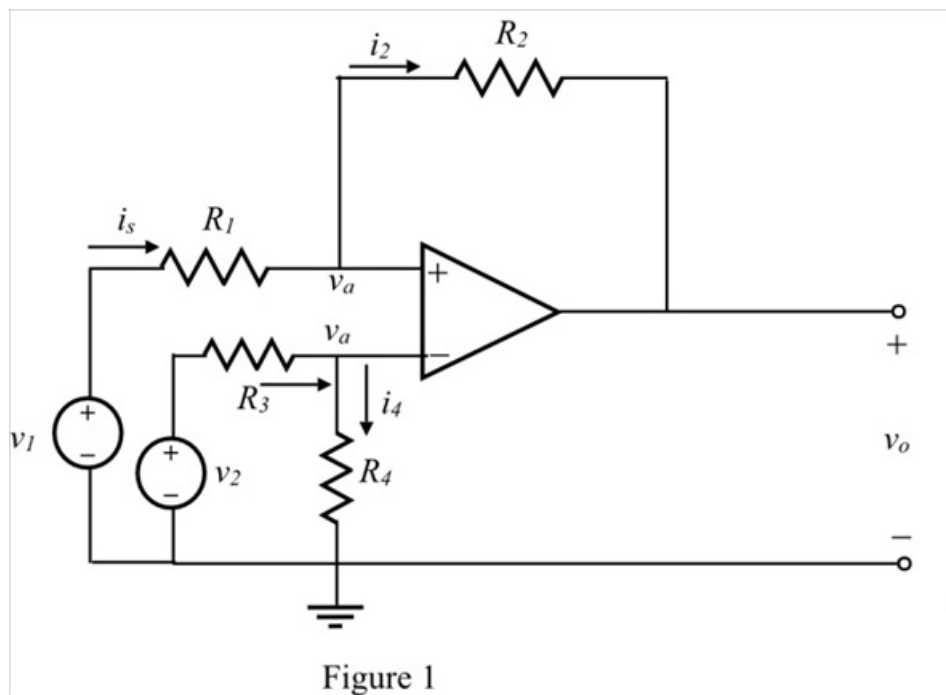
Step-by-step solution

Step 1 of 10

(a)

Refer to Figure 5.24 in the textbook.

Draw the modified circuit diagram as shown in Figure 1.



Step 2 of 10

Determine the voltage across R_4 .

Apply voltage division across R_4 .

$$v_a = \frac{v_2 R_4}{R_3 + R_4} \dots\dots (1)$$

Consider the voltage at non inverting terminal is equal to voltage at inverting terminal.

Apply Kirchhoff's current law at non inverting node.

$$i_s = i_f$$

$$\frac{v_a - v_1}{R_1} + \frac{v_a - v_o}{R_2} = 0$$

$$\frac{v_a}{R_1} - \frac{v_1}{R_1} + \frac{v_a}{R_2} - \frac{v_o}{R_2} = 0$$

$$v_a \left[\frac{1}{R_1} + \frac{1}{R_2} \right] - \frac{v_1}{R_1} = \frac{v_o}{R_2}$$

$$v_o = v_a \left[\frac{R_2}{R_1} + 1 \right] - \frac{v_1 R_2}{R_1} \dots\dots (2)$$

Step 3 of 10

Replace equation 2 in equation 1.

$$\begin{aligned} v_o &= \frac{v_2 R_4}{R_3 + R_4} \left[\frac{R_2}{R_1} + 1 \right] - \frac{v_1 R_2}{R_1} \\ &= \frac{R_2 (1 + R_1 / R_2)}{R_1 (1 + R_3 / R_4)} v_2 - \frac{R_2}{R_1} v_1 \end{aligned}$$

In difference amplifier for $v_1 = v_2$, the output voltage is zero.

$$\frac{R_2 (1 + R_1 / R_2)}{R_1 (1 + R_3 / R_4)} v_1 - \frac{R_2}{R_1} v_1 = 0$$

$$\frac{R_2 (1 + R_1 / R_2)}{R_1 (1 + R_3 / R_4)} v_1 = \frac{R_2}{R_1} v_1$$

$$1 + \frac{R_1}{R_2} = 1 + \frac{R_3}{R_4}$$

$$\frac{R_1}{R_2} = \frac{R_3}{R_4} \dots\dots (3)$$

Step 4 of 10

For difference amplifier the gain is,

$$\frac{v_o}{(v_2 - v_1)} = \frac{R_2}{R_1}$$

Consider the gain of difference amplifier is 2.5.

$$\frac{v_o}{(v_2 - v_1)} = \frac{R_2}{R_1}$$

$$\frac{R_2}{R_1} = 2.5$$

$$R_2 = 2.5R_1$$

Replace $2.5R_1$ for R_2 in equation 3.

$$\frac{R_1}{2.5R_1} = \frac{R_3}{R_4}$$

$$\frac{R_3}{R_4} = \frac{1}{2.5}$$

$$R_4 = R_3(2.5)$$

Step 5 of 10

Consider $R_1 = 20 \text{ k}\Omega$.

Determine the resistance R_2 .

Consider the relation between R_1 and R_2 .

$$\begin{aligned} R_2 &= R_1(2.5) \\ &= 2.5(20 \times 10^3) \\ &= 50 \text{ k}\Omega \end{aligned}$$

Step 6 of 10

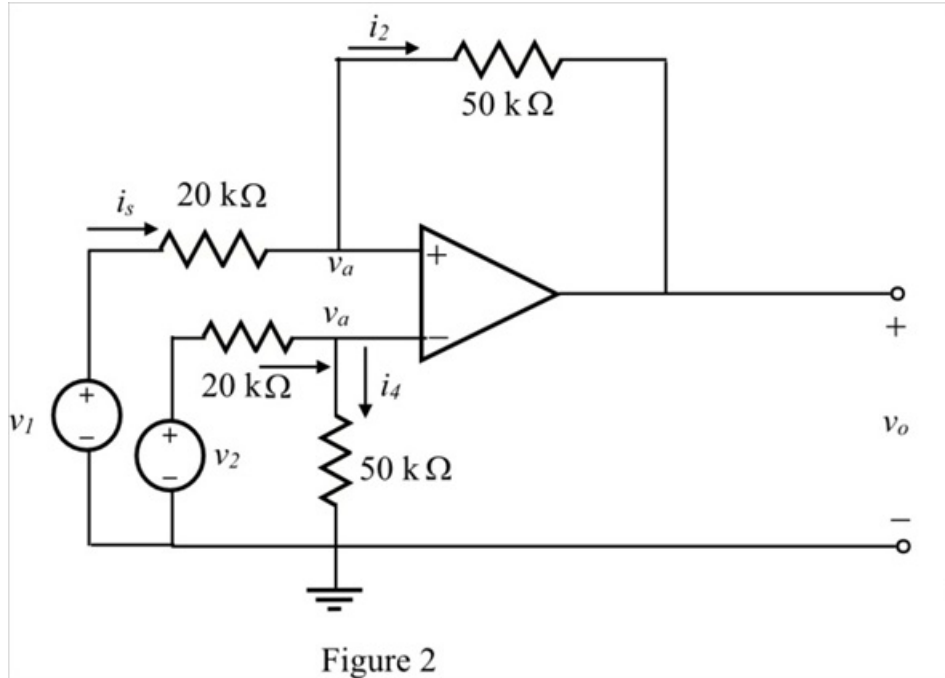
Consider $R_3 = 20 \text{ k}\Omega$.

Determine the resistance R_4 .

Consider the relation between R_3 and R_4 .

$$\begin{aligned} R_4 &= 2.5R_3 \\ &= 2.5(20 \times 10^3) \\ &= 50 \text{ k}\Omega \end{aligned}$$

Replace $20 \text{ k}\Omega$ for R_1, R_3 and $50 \text{ k}\Omega$ for R_2, R_4 in Figure 1 and draw the circuit as shown in Figure 2.

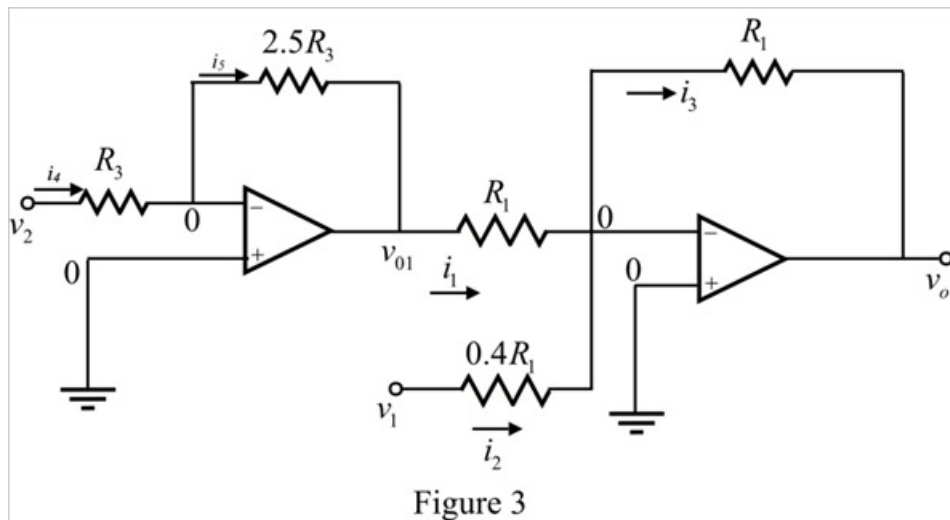


Step 7 of 10

(b)

Consider 2 op-amps are used in designing the difference amplifier.

Draw the circuit as shown in Figure 3.



Step 8 of 10

Apply Kirchhoff's current law at inverting terminal of primary stage.

$$i_4 = i_5$$

$$\frac{0 - v_2}{R_3} + \frac{0 - v_{01}}{2.5R_3} = 0$$

$$\frac{-v_2}{R_3} - \frac{v_{01}}{2.5R_3} = 0$$

$$\frac{v_{01}}{2.5R_3} = \frac{-v_2}{R_3}$$

$$v_{01} = -v_2 (2.5) \dots\dots (4)$$

Apply Kirchhoff's current law at inverting terminal of secondary stage.

$$i_1 + i_2 = i_3$$

$$\frac{0 - v_{01}}{R_1} + \frac{0 - v_1}{0.4R_1} + \frac{0 - v_o}{R_1} = 0$$

$$-\frac{v_{01}}{R_1} - \frac{v_1}{0.4R_1} = \frac{v_o}{R_1}$$

$$v_o = -\left(v_{01} + \frac{v_1}{0.4}\right) \dots\dots (5)$$

Step 9 of 10

substitute equation 4 in equation 5.

$$v_o = -\left(-v_2 (2.5) + \frac{v_1}{0.4}\right)$$

$$v_o = 2.5v_2 - 2.5v_1$$

$$v_o = 2.5(v_2 - v_1)$$

$$\frac{v_o}{(v_2 - v_1)} = 2.5$$

Therefore, the gain of difference amplifier is 2.5.

Step 10 of 10

Consider $R_1 = 10 \text{ k}\Omega$ and $R_3 = 10 \text{ k}\Omega$.

Replace the values in Figure 3 and draw the circuit as shown in Figure 4.

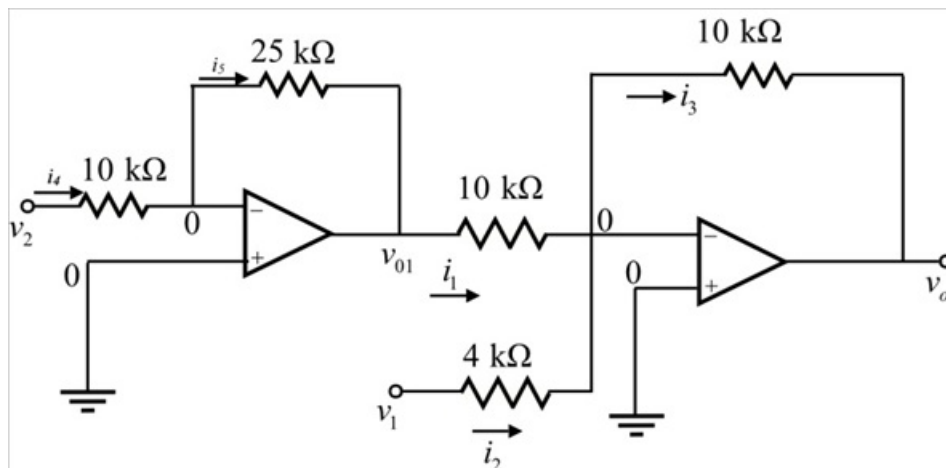


Figure 4